
UNDERSTANDING THE FINE-GRAINED KNOWLEDGE CAPABILITIES OF VISION-LANGUAGE MODELS

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ABSTRACT

Vision-language models (VLMs) now perform strongly across many visual question answering benchmarks, including settings that emphasize visual reasoning, document understanding, and multimodal dialogue. These gains appear across VLM families that differ in base models, alignment architectures, and training data, suggesting broad progress in multimodal interaction. Yet recent work indicates that the same systems can lag on traditional image classification benchmarks that probe fine-grained visual knowledge. We evaluate a broad set of recent VLMs on fine-grained classification benchmarks and analyze factors that may explain the mismatch between fine-grained recognition and commonly used VLM benchmarks. Across a series of controlled ablations, we find that stronger LLMs tend to improve fine-grained and general benchmark scores similarly, whereas stronger vision encoders yield especially large gains on fine-grained classification. We also find that the pretraining stage is important for fine-grained performance, particularly when language model weights are unfrozen during pretraining. Together, these results frame fine-grained recognition as a broadly relevant component of vision-centric VLM capability and point to model and training choices that can improve it.

1 INTRODUCTION

Vision-language models (VLMs), which combine vision encoders with large language models (LLMs), have become increasingly capable across many forms of visual interpretation and reasoning (Alayrac et al., 2022; Team et al., 2023; OpenAI, 2023; Liu et al., 2023; Wang et al., 2024b). Their reported strengths span open-ended visual question answering, document understanding, and multimodal dialogue (Goyal et al., 2017; Yue et al., 2024; Mathew et al., 2021; OpenAI, 2023), indicating broad progress on language-mediated visual tasks. This progress raises a more general question about the visual foundations of such systems: how well do VLMs handle fine-grained visual perception, and which design choices improve that capability?

Fine-grained visual recognition—the ability to distinguish between visually similar categories—requires models to use subtle discriminative cues while remaining robust to within-class changes in pose, lighting, and background. Standalone vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) have been widely tested on fine-grained classification benchmarks, but the extent to which newer LLM-based VLMs preserve or extend these visual capabilities remains less clear.

Understanding fine-grained visual knowledge in VLMs matters beyond any single classification dataset because perception constrains downstream understanding and reasoning. Many practical deployments involve long-tailed visual distributions, where the difference between closely related categories can determine whether later language-based reasoning is useful. For example, a model that misidentifies a mushroom species cannot reliably use textual knowledge to infer toxicity or safe consumption. More generally, identifying which model components and training stages preserve fine-grained information can guide VLM architectures and training strategies for vision-centric use cases.

We first evaluate fine-grained visual knowledge in contemporary VLMs and compare it with performance on broader VLM benchmarks. Concretely, we test 15 VLMs, including LLaVA (Liu et al., 2023; 2024a), Phi (Abdin et al., 2024), Qwen2-VL (Wang et al., 2024b), and Molmo (Deitke et al.,

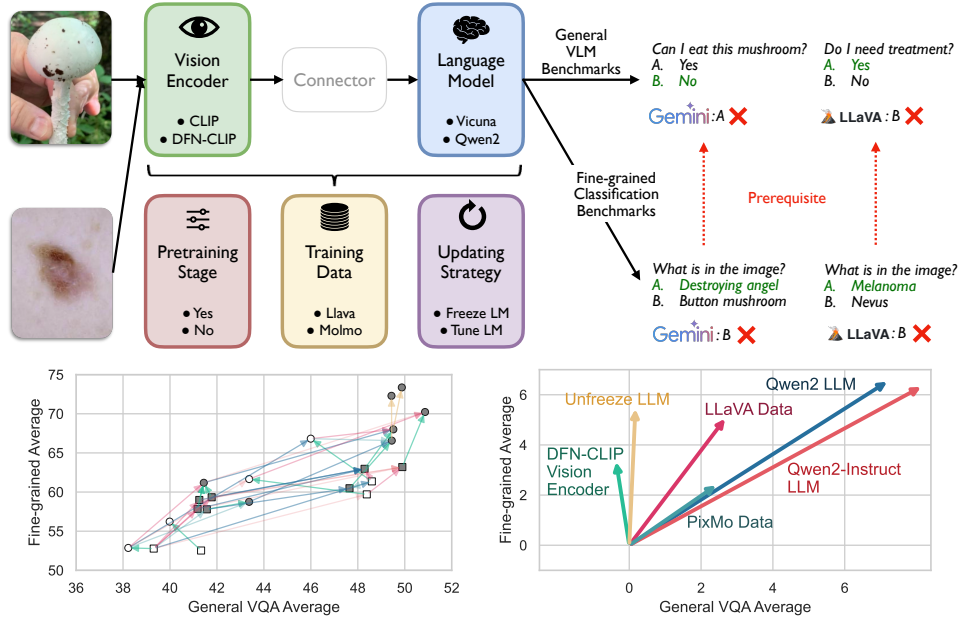


Figure 1: **Overview.** (Top) We study fine-grained classification in vision-language models (VLMs) as a vision-centric capability that supports broader multimodal understanding and reasoning. (Bottom) Across 22 systematic ablations over model components and training strategies (left), we identify factors that shape fine-grained classification performance in VLMs (right).

2024), on four established classification benchmarks: ImageNet (Deng et al., 2009), Flowers (Nilsback & Zisserman, 2008), Pets (Parkhi et al., 2012), and Food (Bossard et al., 2014). We then relate these results to aggregated scores from eight general VLM benchmarks, including MMMU (Yue et al., 2024), MathVista (Lu et al., 2024), and MMVet (Yu et al., 2024). The comparison shows substantial variation in fine-grained classification among models with similar general benchmark scores, indicating that fine-grained recognition is a distinct dimension of visual capability not fully captured by current VLM evaluations. We also observe a consistent gap between VLMs and their CLIP-based vision encoders, in line with prior findings (Zhang et al., 2024; Geigle et al., 2024).

To understand which choices generalize across these fine-grained settings, we conduct a systematic study of model components and training procedures. We ask how the language model, vision encoder, pretraining stage, parameter-updating strategy, and data source affect fine-grained classification relative to general VQA performance. Across 22 controlled ablations, several factors emerge as important for fine-grained knowledge acquisition in VLMs:

- **Language model (§4.2):** Improvements in the base language model transfer broadly, enhancing both fine-grained classification and general VLM benchmarks by similar amounts.
- **Vision encoder (§4.2):** A stronger vision encoder yields larger gains for fine-grained classification than for general VLM benchmarks, especially after two-stage training.
- **Pretraining stage (§4.3):** Image-caption pretraining substantially improves fine-grained classification across our settings while producing smaller changes on general VLM benchmarks.
- **Weight updating method (§4.3):** Fine-grained classification improves when pretraining updates both the connector between the vision encoder and LLM and the LLM itself, rather than the connector alone.
- **Data quality (§4.3):** Within the data sources we compare, pretraining and instruction-tuning data quality has limited effect on fine-grained classification, particularly when language model weights remain frozen during pretraining.

Overall, our study positions fine-grained visual perception as an important but under-measured part of VLM evaluation. By connecting benchmark observations with controlled ablations, we identify design choices that affect fine-grained knowledge acquisition across the tested models and datasets.

These findings provide guidance for building more vision-centric VLMs and for deploying them in long-tail scenarios where precise recognition is a prerequisite for reliable multimodal reasoning.

2 RELATED WORK

Vision-language models and their evaluation. Vision-language models (VLMs) form a broad model class that connects visual inputs to language models, often through a LLaVA-style architecture (Liu et al., 2023) with three main components: a vision encoder, a language model, and an MLP connector between them. Across this family, recent systems have achieved strong results on multimodal tasks such as visual question answering (VQA), document understanding, and general reasoning (Liu et al., 2023; Dai et al., 2023; Awadalla et al., 2023; Wang et al., 2024b). Their progress is usually measured with benchmarks such as MMMU (Yue et al., 2024), MathVista (Wang et al., 2024a), and DocVQA (Mathew et al., 2021). These benchmarks cover important capabilities, but they primarily emphasize reasoning and language understanding conditioned on visual inputs, leaving more vision-centric abilities such as object recognition and fine-grained classification less directly tested. Our work addresses this evaluation gap by systematically analyzing fine-grained classification as a broadly relevant component of visual intelligence in VLMs.

Fine-grained visual classification. Fine-grained classification is a long-standing computer vision task that distinguishes visually similar subcategories within broader object types, including bird species (Wah et al., 2011), flowers (Nilsback & Zisserman, 2008), and pet breeds (Parkhi et al., 2012). Such benchmarks have been central to the development and evaluation of vision encoders including ResNet (He et al., 2016), CLIP (Radford et al., 2021), and DINO (Caron et al., 2021). Although these datasets remain important for measuring detailed visual recognition, they are not commonly included in recent VLM evaluation protocols, which tend to prioritize visual question answering. Because many application domains involve long-tail categories and require precise recognition—for example, medical diagnosis, food safety, and species identification—fine-grained classification provides a useful test of VLM usability beyond language-centric reasoning benchmarks.

Evaluation and ablation of VLMs for fine-grained classification. Recent studies have begun to evaluate VLMs on fine-grained classification (Zhang et al., 2024; Geigle et al., 2024), but existing analyses typically cover a limited set of models or datasets. We broaden this empirical view by evaluating a larger collection of VLMs across multiple fine-grained classification benchmarks using a unified multiple-choice format. We further complement this evaluation with controlled ablations that isolate factors affecting fine-grained classification performance.

Several related works study the VLM design space, including model architectures (McKinzie et al., 2024), training strategies (Karamcheti et al., 2024), and multimodal data choices (Gadre et al., 2023; Udandarao et al., 2024; Deitke et al., 2024). Most of these studies focus on general benchmarks where language and reasoning play a large role. Here, we examine how design choices—including base model selection, pretraining method, and the composition of pretraining and instruction-tuning data—affect fine-grained classification, thereby extending VLM analysis toward vision-centric capabilities.

3 EVALUATION: FINE-GRAINED CLASSIFICATION

This section motivates fine-grained classification as a VLM evaluation axis, describes the benchmarks and evaluation protocol we use, and summarizes the main observations from testing 15 existing VLMs.

3.1 MOTIVATION

Fine-grained classification is a foundational computer vision task that tests whether a model can distinguish visually similar subcategories within a broader category, such as bird species or car models (Wei et al., 2021). Benchmarks such as ImageNet (Deng et al., 2009) and Flowers (Nilsback & Zisserman, 2008) have therefore served as standard testbeds for training and evaluating vision encoders. With large-scale supervised and self-supervised learning, encoders such as CLIP (Radford

et al., 2021) and DINO (Caron et al., 2021) reached near- or superhuman performance on challenging fine-grained classification tasks.

In contrast, these traditional recognition benchmarks are not central to most evaluations of vision-language models (VLMs). Recent VLMs such as LLaVA (Liu et al., 2023) show strong ability to interpret visual concepts, but they are usually assessed through visual question answering (VQA), including academic VQA benchmarks such as MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024), or document understanding benchmarks such as DocVQA (Mathew et al., 2021) and TextVQA (Singh et al., 2019). Those benchmarks capture important reasoning and language-processing behaviors, yet they include relatively few fine-grained object categories, leaving detailed vision-centric recognition less directly measured.

Fine-grained classification is relevant to VLMs because many application scenarios contain long-tailed visual categories where precise distinctions precede any useful reasoning. For example, if a blind user asks whether a mushroom is safe to eat, the model must identify the species before applying knowledge about toxicity; confusing a deadly destroying angel with an edible white button mushroom could have severe consequences. Similar dependencies arise in medical diagnosis, where visually similar diseases may imply different treatments, and in autonomous driving, where visually similar traffic signs can require different actions. These examples illustrate why fine-grained recognition should be evaluated as part of the broader reliability profile of VLMs.

3.2 BENCHMARKS

We evaluate VLM fine-grained visual knowledge using four established object-recognition benchmarks that cover different domains of category-level detail: **ImageNet-1K** (Deng et al., 2009), **Oxford Flowers-102** (Nilsback & Zisserman, 2008), **Oxford-IIIT Pet-37** (Parkhi et al., 2012), and **Food-101** (Bossard et al., 2014).

Because these benchmarks contain 37 to 1,000 classes, direct evaluation in the multiple-choice format commonly used for VLM training and evaluation is nontrivial. We therefore follow Geigle et al. (2024) and convert each dataset into 5-way multiple-choice questions. For each image, an OpenCLIP ViT-L/14 model (Schuhmann et al., 2022) selects the four hardest negative labels by image-label cosine similarity; these distractors are then combined with the ground-truth label to form the answer choices. We use OpenCLIP to avoid making the benchmark preferentially difficult for VLMs whose vision encoders are OpenAI-CLIP (Radford et al., 2021), SigLIP (Zhai et al., 2023), or DFN-CLIP (Fang et al., 2023), each of which is trained on different data than the OpenCLIP model. This conversion lets us evaluate VLMs in a familiar answer format while retaining much of the original classification difficulty. For example, CLIP ViT-L/14-336px (Radford et al., 2021) achieves 75.2% accuracy on the original 102-class Flowers-102 benchmark and 78.3% accuracy on our 5-way multiple-choice version.

3.3 TESTING SETUP

We evaluate 15 recent VLMs in the 7B–13B parameter range on the converted 5-choice fine-grained classification benchmarks (Liu et al., 2023; 2024a; Chen et al., 2024a; Abidin et al., 2024; Deitke et al., 2024; Laurençon et al., 2024; Hong et al., 2024; Chen et al., 2024c; Wang et al., 2024b). For each VLM, we use its default VLMEvalKit prompt format (Duan et al., 2024) and score accuracy by exact match against the multiple-choice options. For baseline comparison, we evaluate several CLIP models (Radford et al., 2021; Schuhmann et al., 2022; Fang et al., 2023) with the standard zero-shot procedure, encoding images and class labels (“A photo of a [class]”) with CLIP image and text encoders. The predicted class is the label whose text embedding has the highest cosine similarity to the image embedding.

To test whether existing VLM evaluations reflect fine-grained recognition, we compare VLM performance on our classification benchmarks with the VLMEvalKit average over eight general VQA benchmarks, including MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024).

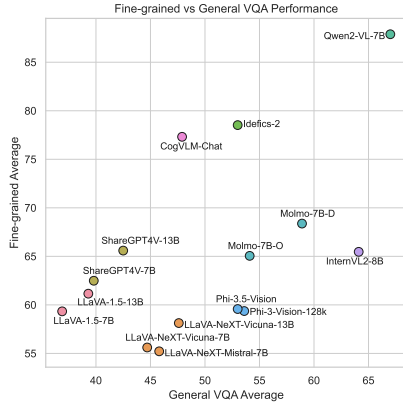


Figure 2: **Fine-grained classification compared to general VQA performance across VLMs.** Across recent VLMs, fine-grained classification varies in ways not explained by aggregate VQA scores, indicating that standard VQA benchmarks do not fully measure this visual capability.

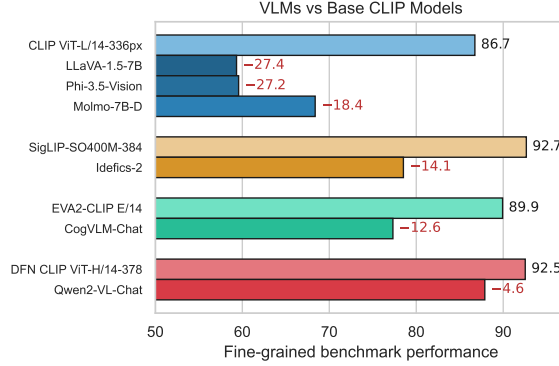


Figure 3: **Comparison of VLMs with their corresponding CLIP vision encoders in fine-grained classification.** Qwen2-VL-Chat nearly matches its DFN-CLIP vision encoder, whereas the other evaluated VLMs remain well below their corresponding encoders, suggesting broadly available headroom for improving fine-grained recognition.

3.4 RESULTS

Figure 2 relates fine-grained classification performance, averaged across our four benchmarks, to general VQA performance, averaged across eight existing VQA benchmarks, for all tested VLMs. Models with similar general VQA scores can differ substantially in fine-grained classification accuracy. For instance, CogVLM-Chat (Hong et al., 2024) and LLaVA-NeXT-Vicuna-13B (Liu et al., 2024a) both achieve nearly 48% average accuracy on general VQA benchmarks, yet their fine-grained classification accuracies differ by 19pp (percentage points): 77.3% for CogVLM-Chat and 58.1% for LLaVA-NeXT-Vicuna-13B. This discrepancy indicates that existing VLM benchmarks do not fully capture fine-grained classification, supporting its use as a distinct evaluation dimension for visual capability.

Finding 1: Fine-grained classification represents a distinct aspect of visual capability that current VLM benchmarks fail to adequately measure.

Because current VLMs are built on pretrained vision encoders, our multiple-choice setup also enables direct comparison between VLMs and their underlying CLIP-style encoders on the same recognition tasks. Figure 3 compares selected VLMs with their corresponding CLIP baselines and shows a substantial performance gap. For example, Molmo (Deitke et al., 2024), a recent model with strong VQA results, obtains 68.4% average accuracy in our multiple-choice evaluation, whereas its CLIP ViT-L/14 (Radford et al., 2021) vision encoder obtains 86.7%. Even the strongest tested VLM, Qwen2-VL-Chat (Wang et al., 2024b), trails its DFN-CLIP (Fang et al., 2023) vision encoder by 4.6 points. These gaps suggest that multimodal integration does not automatically preserve all fine-grained visual knowledge from the encoder and motivate the ablation study in the next section.

Finding 2: A significant performance gap exists between VLMs and CLIP models in fine-grained classification tasks.

4 ABLATIONS: BASE MODELS AND TRAINING

4.1 EXPERIMENTAL SETUP

Building on the observational results in Section 3, we ablate model and training differences that may affect fine-grained classification, with emphasis on base model selection and training method-

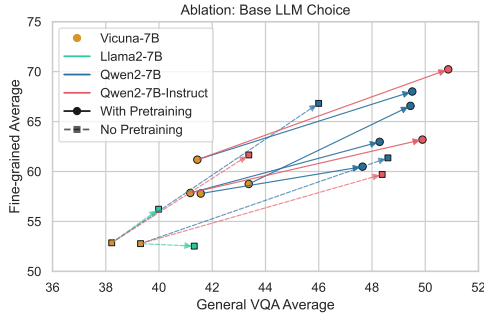


Figure 4: **Ablating the base LLM by replacing Vicuna-7B in LLaVA with other LLMs.** On average, switching from Vicuna to Qwen2-7B yields a +7.5pp gain in fine-grained performance and a +7.7pp gain in general VQA performance, suggesting that LLM improvements transfer across both evaluation types.

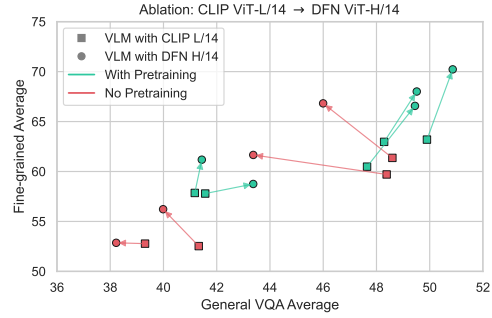


Figure 5: **Ablating the base vision encoder by replacing CLIP ViT-L/14 in LLaVA with DFN-CLIP ViT-H/14.** With sufficient pre-training, DFN-CLIP improves fine-grained performance by +4.5pp and general performance by +1.2pp, showing a larger effect on detailed recognition.

ology. We use the LLaVA-1.5 training framework and alignment architecture (Liu et al., 2023) as a controlled setting in which these choices can be compared. Each model receives one epoch of pretraining when applicable, followed by one epoch of finetuning. Across experiments, we keep LLaVA-1.5’s default hyperparameters, including learning rate and batch size, fixed so that observed changes reflect the ablated factors.

Fine-grained performance is reported as the average score across the four classification benchmarks described in Section 3.2. For general VQA performance, we report the average over three widely used multiple-choice benchmarks: MMMU (Yue et al., 2024), MMBench (Liu et al., 2024b), and MMStar (Chen et al., 2024b). These benchmarks provide domain coverage and are widely adopted in VLM evaluation, giving a compact comparison point for changes that may or may not generalize beyond fine-grained classification.

4.2 BASE MODELS

The analysis in Section 3 shows that recent VLMs often differ in their base LLMs, while many still rely on the CLIP ViT-L/14 vision encoder from Radford et al. (2021). We therefore isolate how these base model choices affect fine-grained classification and whether their effects carry over to general VQA benchmarks.

4.2.1 LLM CHOICE

Setup. Using LLaVA-1.5-7B (Liu et al., 2023) as the baseline, we vary the LLM component across multiple vision encoders and pretraining configurations. We compare the original Vicuna-7B-1.5 (Chiang et al., 2023) used in LLaVA with Qwen2-7B (Yang et al., 2024), which is used in Qwen2-VL-7B (Wang et al., 2024b) and Molmo-7B-D (Deitke et al., 2024), as well as Llama2-7B (Touvron et al., 2023), Vicuna’s base model, and Qwen2-7B-Instruct, the instruction-tuned variant of Qwen2-7B.

Findings. As shown in Figure 4, replacing Vicuna-7B with Qwen2-7B improves performance across all tested configurations, with average gains of +7.5pp (percentage points) on fine-grained benchmarks and +7.4pp on general VQA benchmarks. The instruction-tuned Qwen2-7B-Instruct produces similar improvements, at +7.5pp and +8.1pp respectively. Llama2-7B, the non-instruction-tuned base of Vicuna, provides smaller gains, though the downstream implications of using a model without instruction tuning remain unclear. All trained models, including lower-performing Vicuna-based VLMs, consistently produce multiple-choice responses in the correct format. Thus, the fine-grained differences appear to reflect knowledge of the presented options rather than merely better adherence to answer formatting.

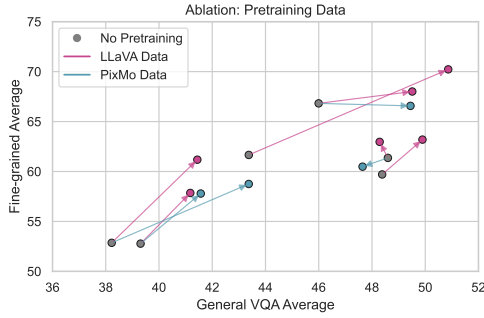


Figure 6: **Ablating pretraining data.** Adding either LLaVA (CC-3M) or Molmo (PixMo-Cap) data produces similar gains. LLaVA data increases fine-grained and general scores by +4.0pp and +2.1pp, while PixMo data increases them by +2.4pp and +2.5pp.

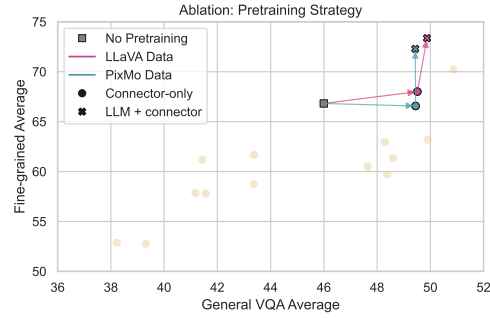


Figure 7: **Ablating training method by comparing connector-only pretraining with LLM+connector tuning.** Unfreezing the LLM during pretraining yields a +5.5pp improvement in fine-grained performance without reducing general VQA scores.

Takeaway 1: Stronger language models consistently improve performance across both fine-grained classification and general VQA benchmarks.

4.2.2 VISION ENCODER CHOICE

Setup. Many evaluated VLMs, including the LLaVA series and Molmo, use OpenAI-CLIP ViT-L/14-336px (Radford et al., 2021) as their vision encoder. Qwen2-VL (Wang et al., 2024b), by contrast, uses DFN-CLIP ViT-H/14-378px (Fang et al., 2023), which has stronger zero-shot performance on object-recognition tasks such as ImageNet. We compare these two vision encoders across several LLM choices and pretraining configurations to determine when improved visual representations benefit VLM outputs.

Findings. Figure 5 shows that the effect of replacing CLIP L/14 with DFN-CLIP H/14 depends on whether connector pretraining occurs before finetuning. Without pretraining, DFN-CLIP reduces general VQA performance (-2.5pp) while modestly improving fine-grained scores (+2.8pp). After connector pretraining, however, general VQA performance remains stable or slightly improves (+1.2pp), and fine-grained performance increases more substantially (+4.5pp). This pattern indicates that stronger vision encoders can improve detailed recognition in VLMs, but the benefit is realized more reliably when the encoder is integrated through pretraining.

Takeaway 2: Better vision encoders (like DFN-CLIP vs. CLIP) in VLMs improve fine-grained classification, but only when these encoders are properly integrated through pretraining before the finetuning stage.

4.3 TRAINING METHODS

Recent VLMs are commonly trained in two stages (Liu et al., 2023; Wang et al., 2024b; Chen et al., 2024c). The first stage, pretraining, uses image-caption data and a captioning objective to align the vision encoder with the LLM. The second stage, instruction tuning, uses multi-round multimodal instruction-completion data to adapt the model toward user-facing tasks. We examine how these stages affect fine-grained capabilities and whether their effects extend similarly to general VQA benchmarks.

4.3.1 PRETRAINING

Setup. We investigate three questions about the pretraining process:

- Is pretraining necessary for LLaVA-architecture models when the base models are already pretrained on billion-scale data?

Table 1: **Summary of key ablation settings and comparison to existing models (bolded)**. Across the tested settings, upgrading the base LLM and vision encoder and pretraining the connector and LLM improve fine-grained performance, while instruction finetuning has a smaller effect. For general VQA benchmarks, the LLM choice accounts for a larger share of the observed gains.

Model/Ablation	Vision Encoder	LLM	Arch.	Pretraining Data	Finetuning Data	Fine-grained Classification	General VQA
LLaVA-1.5-7B	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	59.3	41.8
No pretraining	CLIP L/14	Vicuna	LLaVA	None	LLaVA	52.8	39.3
LLaVA reproduction	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	57.9 (+5.1)	41.2 (+1.9)
Qwen2 LLM	CLIP L/14	Qwen2	LLaVA	LLaVA	LLaVA	63.0 (+5.1)	48.3 (+7.1)
DFN-CLIP encoder	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	68.0 (+5.0)	49.5 (+1.2)
Unfreeze LLM	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	73.4 (+5.4)	49.9 (+0.4)
Pretrain on PixMo	CLIP L/14	Qwen2	LLaVA	PixMo	LLaVA	66.6	49.4
Molmo-7B-D	CLIP L/14	Qwen2	Molmo	PixMo	Molmo	68.4	58.0
FT Qwen2-VL base on LLaVA	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	LLaVA	85.5	61.1
Qwen2-VL-7B	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	Qwen2-VL	87.9 (+2.4)	62.4 (+1.3)

- How does connector-only pretraining compare with unfreezing the LLM?
- How important is pretraining data quality for downstream performance?

We compare three pretraining configurations: LLaVA pretraining data (Liu et al., 2023), a subset of CC-3M with web-scraped captions; Molmo pretraining data (Deitke et al., 2024), PixMo-Cap with detailed human-annotated captions; and no pretraining. Following Liu et al. (2023), most experiments pretrain only the connector. We also test a strategy similar to Deitke et al. (2024): train the connector alone for 20% of the steps, then unfreeze the LLM for the remaining 80%. Every configuration is then finetuned for one epoch on LLaVA finetuning data.

Findings. Figure 6 summarizes the pretraining ablations. Connector-only pretraining generally improves VLM performance, except for the Qwen2 and CLIP ViT-L/14 combination. The gains are especially large for Vicuna-based VLMs, where average performance increases by +6.1pp on fine-grained classification and +3.1pp on general VQA.

Takeaway 3: Large-scale pretraining on image captioning datasets substantially improves fine-grained classification performance but has a more modest effect on general VLM benchmarks.

Within our training setup, pretraining on web-scraped captions (LLaVA) and on detailed human annotations (PixMo) leads to only small differences. Compared with LLaVA data, PixMo data changes fine-grained performance by -1.6pp and general VQA performance by +0.4pp.

Takeaway 4: Pretraining data quality has a limited impact on overall model performance.

One possible explanation is that connector-only pretraining limits how much the model can use richer captions. To test this, we pretrain both the LLM and connector after a 20% connector-only warmup phase. Figure 7 shows that this strategy improves fine-grained classification by +5.5pp without reducing general VQA scores. Because the improvement appears with both LLaVA and PixMo pretraining data, the result suggests that, in this setting, lower-quality captions do not substantially impair the LLM’s language capabilities.

Takeaway 5: Pretraining both LLM and connector substantially enhances fine-grained benchmark performance while maintaining general VQA scores.

4.3.2 FINETUNING

Setup. For this ablation, we move beyond the previous LLaVA-style setup and start from the pre-trained Qwen2-VL-7B-Base model (Wang et al., 2024b), the base model used to train Qwen2-VL-7B-Chat. We finetune it on LLaVA-Instruct data (Liu et al., 2023) using our earlier protocol and compare the result with Qwen2-VL-7B-Chat, which uses 2M in-house examples for instruction finetuning (Wang et al., 2024b). This comparison isolates the effect of the instruction-finetuning data source within the Qwen2-VL model family.

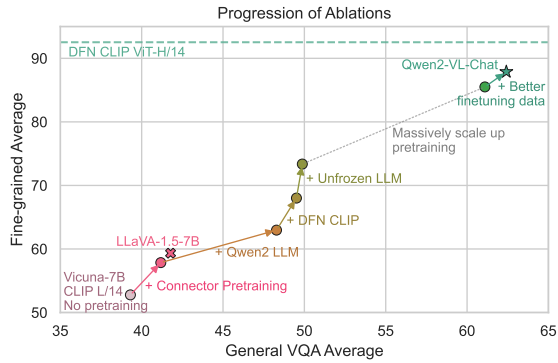


Figure 8: **Progression of combined ablation effects.** Starting from Vicuna-7B and CLIP ViT-L/14, successive changes to the base LLM, vision encoder, and training settings raise fine-grained classification accuracy from 52.8% to 73.4%, explaining more than half of the gap to Qwen2-VL-Chat’s 87.9%. We attribute the remaining difference primarily to Qwen2-VL’s much larger 1.4T-token pretraining corpus.

Findings. Finetuning only on LLaVA data slightly lowers both fine-grained and general benchmark performance. Fine-grained scores decrease by 2.4pp, from 87.9% to 85.5%, while general benchmark scores decrease by 1.3pp, from 62.4% to 61.1%. As shown in Figure 8, this is the smallest fine-grained contribution among our ablations, suggesting that instruction finetuning data has less influence on fine-grained knowledge than base model selection and pretraining strategy in the settings we test.

Takeaway 6: The instruction finetuning stage has comparatively less impact on fine-grained classification performance than other factors.

4.4 SUMMARY OF ABLATIONS

Figure 8 and Table 1 summarize how the ablated components reduce the gap between weaker and stronger VLMs on fine-grained and general VQA tasks. Starting from the LLaVA (Liu et al., 2023) with Vicuna-7B (Chiang et al., 2023) and CLIP ViT-L/14 (Radford et al., 2021), we modify components step by step, increasing fine-grained performance from 52.8% to 73.4% and general VQA performance from 39.3% to 49.9%. The pattern across these changes shows that switching to DFN-CLIP (Fang et al., 2023) and unfreezing the LLM during pretraining benefit fine-grained capability more than general VQA, whereas switching the LLM to Qwen2-7B (Yang et al., 2024) explains most of the general VQA gain.

Despite these improvements, Table 1 shows that a 12-point fine-grained classification gap remains outside the factors directly covered by our ablations. We consider two plausible contributors to this remaining difference: architectural variation and pretraining data scale.

Architectural differences may matter, but our observational results in Section 3.4 and training ablations in Section 4.3 suggest that pretraining data scale is likely a major factor. The LLaVA (Liu et al., 2023) and PixMo (Deitke et al., 2024) datasets used in our experiments are relatively small, each containing fewer than 1M images and captions, or approximately 200M and 400M tokens, respectively. By contrast, Wang et al. (2024b) report pretraining Qwen2-VL on 1.4T tokens, orders of magnitude more than the scale used here. This scale difference likely explains part of the remaining performance gap and points to extensive pretraining as an important ingredient for VLMs with stronger fine-grained classification.

5 LIMITATIONS & CONCLUSION

Our study has practical limitations that define the scope of its conclusions and suggest directions for future work. Because of computational constraints, our controlled training comparisons use <1M data points rather than the billion-scale training regimes used by some newer VLMs, so our findings

characterize moderate-scale ablations rather than the full scaling behavior of VLM pretraining. In addition, recent work proposes alternative pretraining strategies that may interact differently with fine-grained visual understanding and should be tested in future studies.

In this work, we systematically evaluate state-of-the-art vision-language models (VLMs) on fine-grained classification benchmarks and show that fine-grained visual classification is a distinct, under-measured dimension of VLM capability. Through controlled analysis of model components and training paradigms, we identify factors that improve fine-grained recognition within the tested settings and clarify how those factors differ from the ones that drive general VQA performance. These results support a broader view of VLM evaluation in which precise visual understanding complements language-mediated reasoning for applications that require reliable recognition.

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Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	Classification	VLM
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.3	41.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.0	41.2
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.8	39.3
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	52.5	41.3
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	56.2	40.0
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.9	38.2
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.4	48.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	66.8	46.0
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	59.7	48.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	61.7	43.4
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	85.5	61.1
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.9	41.2
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	61.2	41.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	63.0	48.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	68.0	49.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	63.2	49.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	70.2	50.9
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	87.9	62.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	41.6
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	58.8	43.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	60.5	47.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	66.6	49.4
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	73.4	49.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	72.3	49.4

Table 2: **Detailed aggregate results for the model ablations.** The table reports how the tested combinations of vision encoder, LLM, architecture, pretraining data, finetuning data, and trainable parameters affect both fine-grained and general VQA averages.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	MMBench	MMMU	MMStar
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	32.2	32.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	58.6	32.9	32.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	54.4	32.8	30.7
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	58.0	34.3	31.7
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	53.9	34.6	31.5
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	51.3	33.2	30.1
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	63.1	41.8	40.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.3	36.2	40.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	63.2	41.4	40.5
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	55.7	38.3	36.1
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	78.5	49.0	55.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	29.9	33.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.4	34.2	32.7
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.4	41.3	39.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.9	41.3	42.3
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	66.5	41.3	41.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	68.0	41.6	43.0
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	78.9	49.9	58.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	32.6	34.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	61.1	34.1	34.9
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	64.9	40.0	38.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	63.0	41.0	44.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	66.0	42.0	41.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	63.7	40.0	44.6

Table 3: **Detailed general VQA results for the model ablations.** Scores on MMBench, MMMU, and MMStar show how each ablated configuration transfers to broader multiple-choice VLM benchmarks.

A FULL RESULTS

We report the full ablation results for aggregate performance, general VQA benchmarks, and fine-grained classification in Table 2, Table 3, and Table 4, respectively, to show how the observed patterns hold across the individual evaluation slices.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	ImageNet	Flowers	Pets	Food
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	70.9	50.7	46.3	69.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	68.7	50.4	46.5	70.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	64.3	46.1	40.4	60.4
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	64.4	44.2	40.7	60.8
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	65.4	50.5	46.2	62.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	62.3	48.7	39.2	61.3
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	70.6	44.1	55.6	75.2
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	74.1	50.8	62.3	80.1
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	71.4	39.6	53.1	74.7
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	72.0	46.0	54.1	74.5
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	84.8	79.5	87.1	90.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	69.1	50.0	45.3	67.1
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	72.7	53.2	48.5	70.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	71.5	46.9	56.6	76.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	76.1	54.1	59.1	82.7
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	74.5	45.8	55.1	77.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	78.5	52.3	65.4	84.7
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	85.9	82.3	91.0	92.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	69.2	47.4	46.7	67.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	70.8	50.9	44.8	68.5
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	70.8	43.7	53.9	73.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	76.1	50.3	57.8	82.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	79.4	60.7	67.3	86.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	80.0	56.4	65.4	87.4

Table 4: **Detailed fine-grained classification results for the model ablations.** Scores on ImageNet, Flowers, Pets, and Food show how each ablated configuration affects recognition across the evaluated fine-grained domains.

B EVALUATION

B.1 FINE-GRAINED CLASSIFICATION

As described in Section 3.2, we evaluate VLMs and their base vision encoders on four traditional classification benchmarks that cover complementary fine-grained recognition domains:

ImageNet-1K (Deng et al., 2009): A foundational image classification dataset organized around the WordNet hierarchy, covering a wide range of supercategories that include many fine-grained subcategories. The ImageNet test set contains 50,000 images spanning 1,000 classes.

Because of the ImageNet/WordNet class hierarchy, some classes are too visually distinct from the rest of the label set to support hard multiple-choice distractors; for example, only three classes fall under the “person” category. To address this issue, and to keep the evaluation size manageable, we use human-annotated incorrect model predictions from Recht et al. (2019) to select images for which hard negative choices can be constructed. From 19,056 annotated examples, we retain 5,457 images covering 928 of the original 1,000 classes. Because this curation produces an uneven class distribution, we report mean-per-class accuracy for ImageNet.

Oxford Flowers-102 (Nilsback & Zisserman, 2008): A 102-species flower dataset with substantial intra-class variation and inter-class similarity. The Flowers test set contains 6,149 examples.

Oxford-IIIT Pet-37 (Parkhi et al., 2012): A 37-category pet dataset that includes variation in pose, lighting, and occlusion. The Pets test set contains 3,669 examples.

Food-101 (Bossard et al., 2014): A 101-category food dataset with substantial visual diversity within each class. The Food test set contains 25,250 examples.

Question generation. For all fine-grained classification test sets except ImageNet, which is curated as described above, we generate hard negatives by running zero-shot classification with an OpenCLIP ViT-L/14 trained on LAION (Schuhmann et al., 2022), following Geigle et al. (2024). We take the top 4 incorrect predictions and shuffle them with the correct label to form the answer choices.

Prompt formatting. In our multiple-choice VQA format, the question is always “What type of object is in this photo?” The answer choices are formatted as “A. Option 1\n B. Option 2\n C. Option

3\n D. Option 4\n E. Option 5”. Because VLMs are trained with different multiple-choice prompts and formatting conventions, we use each model’s corresponding VLMEvalKit format (Duan et al., 2024); one example is “Answer with the option’s letter from the given choices directly.”

For CLIP models, we use the standard zero-shot evaluation procedure, so no question prompt is used. Class names are passed to the text encoder with the template “a photo of a {classname}”.

B.2 GENERAL VQA

For the initial observational analysis in Section 3, we use the scores reported on the OpenVLM Leaderboard (Duan et al., 2024). The default leaderboard average spans eight diverse VLM benchmarks: MMBench (Liu et al., 2024b), MMStar (Chen et al., 2024b), MMMU (Val) (Yue et al., 2024), MathVista (Lu et al., 2024), OCRBench (Liu et al., 2024c), AI2D (Kembhavi et al., 2016), HallusionBench (Guan et al., 2024), and MMVet (Yu et al., 2024). We use this reported average only for Figure 2.

For subsequent evaluations, including existing models and the models trained in Section 4, we restrict the general VQA comparison to multiple-choice benchmarks so the format is more directly comparable. Specifically, we evaluate VLMs on three benchmarks:

MMBench (Liu et al., 2024b): A 2,948-example dataset covering a broad range of VLM capabilities split between perception and reasoning. Most examples involve coarse-grained perception or visual reasoning, although some are labeled as “fine-grained perception”; those categories, such as action recognition, attribute recognition, and OCR, differ from the fine-grained classification domains studied here.

MMMU (Val) (Yue et al., 2024): A large-scale multimodal dataset covering a broad distribution of domains and applications. Although the test set contains 10,500 examples, most works report performance on the 900-question validation split. MMMU evaluates general knowledge grounded in visual perception across art, science, engineering, medicine, business, and humanities.

MMStar (Chen et al., 2024b): A 1,500-question dataset selected by human annotators. The examples are chosen to cover diverse tasks across domains and low-level model capabilities.

Evaluation procedure. For the general VQA benchmarks, we use the VLMEvalKit evaluation code (Duan et al., 2024) throughout. Existing models follow their predefined evaluation procedures, and our trained models use the LLaVA evaluation code because they are finetuned exclusively on LLaVA-Instruct data.

C TRAINING

We build on the LLaVA-1.5 training codebase (Liu et al., 2023). For most experiments, we keep the LLaVA architecture and training procedure fixed so that changes in performance can be attributed to the ablated components.

Pretraining. Pretraining consists of one epoch on either the LLaVA pretraining data, derived from CC-3M images and captions, or the Molmo pretraining data, PixMo-Cap. For PixMo-Cap, we use only the “caption” field and exclude the raw audio transcript. The pretraining batch size is 256. In most pretraining runs, we update only the randomly initialized MLP connector with a learning rate of $1e-3$. When ablating the pretraining strategy, we first train only the connector for a 20% warmup, then unfreeze the LLM and train it with a learning rate of $2e-5$ for the remaining 80% of steps.

Finetuning. For finetuning, every model is trained for one epoch on the LLaVA-1.5 finetuning data, which combines LLaVA-Instruct with VQA training data from several datasets to support benchmarking. During finetuning, the LLM learning rate is $2e-5$ and the batch size is 128. The vision encoder remains frozen in both training phases.

For the ablation on the Qwen2-VL architecture, we start from Qwen2-VL-7B Base and use the same method, unfreezing only the LLM.

Compute. We train all models on 4 nodes with 4 A100 (40GB) GPUs per node. Training usually finishes within 20 hours, with runtime varying by CLIP model size.

LLaVA vs. PixMo data. The main text compares LLaVA and PixMo pretraining data as two caption sources with different annotation characteristics. LLaVA pretraining data comes from CC-3M, which contains web-scraped captions that are often short and only loosely aligned with image content. PixMo-Cap, by contrast, is collected by having human annotators record 60–90 second audio descriptions of images, which are then converted into image captions (Deitke et al., 2024). A surface analysis shows that LLaVA captions average 9.8 words, whereas PixMo captions average 169 words.

LLaVA caption example: “front panel bracket cover for suzuki”

PixMo caption example: “The poster from the TV show ”Law and Order” features a past cast ensemble set in an interrogation room. On the left is a Hispanic man, identified as Detective Nick Amaro, portrayed by Danny Pino. He has dark hair, olive skin, and is dressed in a gray jacket, blue shirt, and black tie. Standing beside him is a blonde woman with shoulder-length hair, who is identified as Detective Amanda Rollins, played by Kelli Giddish. She is ... The backdrop is an interrogation room, with a concrete wall and a portion of a two-way mirror visible, adding to the procedural drama’s atmospheric setting.”